

# An Enhanced Market Maker for the Bristol Stock Exchange Using Bayesian Optimisation and EMA Trend Detection

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**Abstract.** This paper presents MMM02, an improved automated market maker for the Bristol Stock Exchange (BSE), built upon the baseline MMM01 strategy. Since MMM01 relies on fixed buying thresholds with no awareness of market conditions or inventory, MMM02 addresses these limitations through three enhancements: an Exponential Moving Average (EMA) crossover for detecting uptrends or downtrends, volatility-adaptive thresholds - to adapt for more stable or unstable markets - and inventory management to prevent the agent from being exposed to excessive losses by tracking how long a unit has been held for. Hyperparameters are tuned via Bayesian optimisation on a training dataset and the final configuration is tested on SPY intraday data from 3rd March 2026. MMM02 attained a significantly higher mean final balance than MMM01 (2798 vs 915 cents, a 206% improvement), with substantially greater profit per trade (52.73 vs 7.44 cents) despite executing fewer trades on average. This is confirmed through the Mann-Whitney U test and Wilcoxon signed-rank test (both  $p < 0.0001$ ).

**Keywords:** Market Making · Bristol Stock Exchange · EMA Crossover · Bayesian Optimisation · Inventory Management

## 1 Introduction

BSE is a simplified simulation of a limit order book exchange and its matching engine, providing a controlled environment in which automated trading algorithms can be developed and evaluated [1]. Robot traders interact through a continuous double auction, with market makers acting as proprietary traders that use their own capital to profit from intraday price movements.

The baseline market maker, MMM01, follows a simple buy-low sell-high strategy which involves purchasing when the ask falls below 85% of the recent average and selling at a fixed 25 cent profit margin. However, MMM01 is not aware of market direction, applying the same thresholds regardless of the direction and volatility of the market.

This paper presents MMM02, an enhanced market maker that directly addresses these limitations using three key modifications: an EMA crossover signal to identify market trends, volatility-adaptive thresholds that adjust the buying

aggressiveness based on the stability of the market, and inventory management that limits exposure to loss when a unit has been held for an extended duration.

## 2 MMM02 Design

### 2.1 Exponential Moving Average Trend Detection

An EMA is a weighted average of recent values that weights recent observations more heavily, making it more responsive to current market conditions than a simple moving average [2]. For each new transaction price  $p$ , the EMA is updated as:

$$\text{EMA} = \alpha \cdot p + (1 - \alpha) \cdot \text{EMA}_{\text{prev}} \quad (1)$$

where  $\alpha = \frac{2}{w+1}$ , and  $w$  is the window size in number of trades.

MMM02 uses two EMAs - a fast EMA with window ( $w_{\text{fast}}$ ) and a slow EMA with window ( $w_{\text{slow}}$ ) - generating a trend signal via a crossover mechanism. When  $\text{EMA}_{\text{fast}} > \text{EMA}_{\text{slow}}$ , an uptrend is detected and MMM02 is encouraged to buy; when  $\text{EMA}_{\text{fast}} < \text{EMA}_{\text{slow}}$ , a downtrend is detected and MMM02 switches to sell mode, offloading any held inventory.

### 2.2 Volatility-Adaptive Thresholds

While the buy threshold and the profit target for MMM01 are static, MMM02 adapts to the price dynamics using a rolling standard deviation  $\sigma$  over a fixed window of the last 30 transaction prices. The adaptive buy threshold  $\theta_{\text{buy}}$  and profit target  $\delta_{\text{sell}}$  are computed as:

$$\theta_{\text{buy}} = \theta_{\text{base}} - \lambda \cdot \frac{\sigma}{\bar{p}} \quad (2)$$

$$\delta_{\text{sell}} = \delta_{\text{base}} + \lambda \cdot \sigma \quad (3)$$

where  $\theta_{\text{base}}$  and  $\delta_{\text{base}}$  are the base buy threshold and base profit target respectively,  $\bar{p}$  is the mean of recent transaction prices, and  $\lambda$  controls the sensitivity of the adaptation. With higher volatility, buy level  $\theta_{\text{buy}}$  decreases, requiring a larger discount, and  $\delta_{\text{sell}}$  increases to compensate for elevated risk. Both parameters are clipped to  $[0.50, 0.99]$  and  $[5, 100]$  cents respectively, to ensure trading capacity in all scenarios.

### 2.3 Inventory Management

MMM01 holds at most one unit at a time, alternating between buying and selling, which limits its ability to capitalise on prolonged uptrends as it must sell before buying again. MMM02 introduces an inventory management system that supports holding up to a maximum of  $K$  units simultaneously. Purchase prices are tracked in a first-in first-out (FIFO) queue, with the oldest purchase price

being used when calculating the profit target for each sale. This allows MMM02 to accumulate units during detected uptrends before selling them. In downtrend conditions, MMM02 reduces its profit target to half of  $\delta_{\text{sell}}$ , encouraging faster liquidation of held inventory to avoid holding assets in a falling market.

## 2.4 Hyperparameter Optimisation

MMM02 is controlled by six hyperparameters which determine its trading behaviour. Instead of selecting them manually, Bayesian optimisation with a Tree-structured Parzen Estimator (TPE) sampler [3] was used to efficiently search the parameter space. The TPE sampler models the objective function using a probabilistic approach and suggests new parameter combinations that are likely to perform well. The objective function was defined as the mean final balance of MMM02 across 3 BSE sessions on the training data, with the optimisation running for 20 trials to reduce the noise in the objective evaluation. The hyperparameters and their search ranges are summarised in Table 1.

**Table 1.** MMM02 hyperparameters and their search ranges.

| Parameter                                    | Search Range |
|----------------------------------------------|--------------|
| Fast EMA window $w_{\text{fast}}$            | 3 - 25       |
| Slow EMA window $w_{\text{slow}}$            | 15 - 60      |
| Max inventory $K$                            | 1 - 5        |
| Buy threshold $\theta_{\text{base}}$         | 0.80 - 0.99  |
| Profit target $\delta_{\text{base}}$ (cents) | 3 - 50       |
| Volatility sensitivity $\lambda$             | 0.5 - 5.0    |

## 3 Experimental Setup

Two SPY datasets were prepared: intraday prices from 2nd March 2026 for hyperparameter tuning, and intraday prices from 3rd March 2026 for testing the final configuration. Both datasets had one-minute intervals with price offsets normalised and scaled by a factor of 75.

For each trading session, there were 12 regular traders, three for each of ZIP, SHVR, GVWY, and SNPR sellers, and ZIP, ZIC, SHVR, and GVWY buyers. In addition, there was one trader each for MMM01 and MMM02. A total period of 7.5 hours was simulated for each trading session. Customer orders were generated by sending order requests to the regular traders via a stochastic process, where each set of orders was sent within a span of 30 seconds with random inter-arrival intervals.

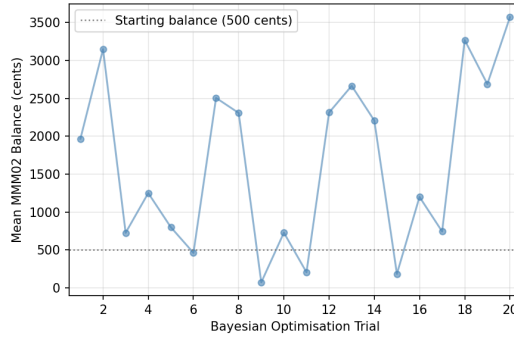
Both systems, MMM01 and MMM02, traded in the same market session for 30 independent trials, hence any difference between their results could be attributed to differences in strategies rather than variation in market conditions. Both market makers were assigned an initial balance of 500 cents. Three metrics

were used: the final balance to measure overall profitability, number of trades to capture trading activity and profit per trade to measure efficiency.

## 4 Results

### 4.1 Optimal Hyperparameters

The objective value for each Bayesian optimisation trial is illustrated in Figure 1, exhibiting considerable variance across trials, reflecting the stochastic nature of the BSE sessions. The strongest performing configuration was found in the final trial.



**Fig. 1.** Bayesian optimisation objective values over the 20 trials. Each point represents the mean MMM02 balance achieved by a single hyperparameter configuration.

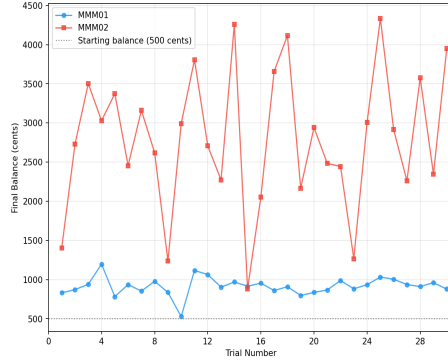
**Table 2.** Optimal MMM02 hyperparameter values found by the Bayesian optimisation.

| Parameter                                    | Optimal Value |
|----------------------------------------------|---------------|
| Fast EMA window $w_{\text{fast}}$            | 13            |
| Slow EMA window $w_{\text{slow}}$            | 51            |
| Max inventory $K$                            | 1             |
| Buy threshold $\theta_{\text{base}}$         | 0.898         |
| Profit target $\delta_{\text{base}}$ (cents) | 31            |
| Volatility sensitivity $\lambda$             | 0.709         |

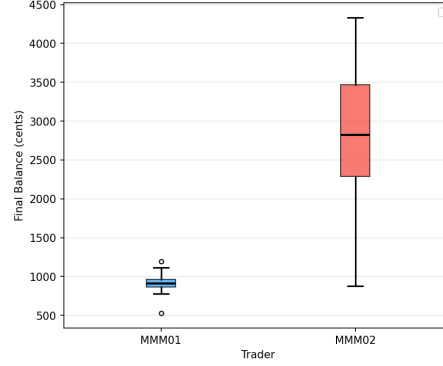
The ideal window sizes for the EMA strategies, which are 13 and 51, strike a balance between sensitivity and resilience to short-term fluctuations. A position size of one was chosen despite the search range being up to five, suggesting that holding one unit is sufficient and accumulating multiple positions does not improve performance.

## 4.2 Performance Comparison

Figure 2 depicts MMM02 outperforming MMM01 in nearly all 30 trials. Figure 3 validates this, where MMM02’s distribution is much greater than MMM01’s with minimal overlap, though with considerably greater variance reflecting MMM02’s sensitivity to market conditions in each trial.



**Fig. 2.** Trial-by-trial final balance comparison across 30 independent test sessions.



**Fig. 3.** Distribution of final balances over 30 trials for MMM01 and MMM02.

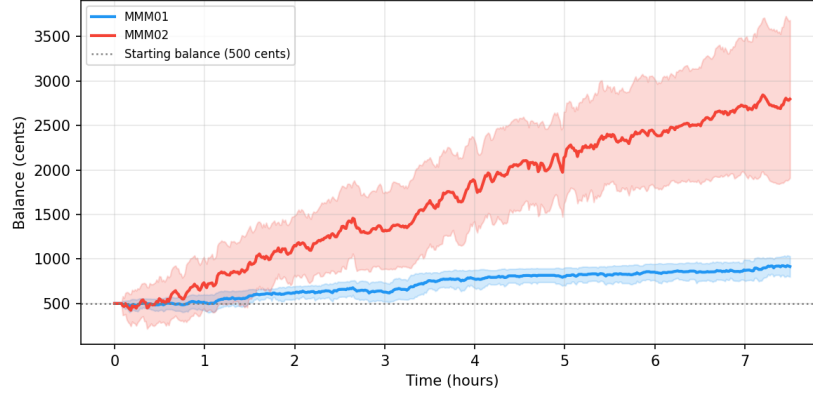
The statistics in Table 3 confirm the above findings. MMM02 recorded an average final balance of 2798 cents compared to 915 cents of MMM01, which was an improvement of 206%. MMM02’s worst trial (880 cents) still approached MMM01’s mean, whilst its best reached 4335 cents. MMM02 executed fewer trades on average (104 vs 142 per session) yet generated substantially greater profit per trade (52.73 vs 7.44 cents), suggesting MMM02 prioritises trade quality over quantity.

**Table 3.** Descriptive statistics for MMM01 and MMM02 across 30 trials.

| Metric                   | MMM01  | MMM02  |
|--------------------------|--------|--------|
| Mean balance (cents)     | 915.4  | 2798.2 |
| Std deviation (cents)    | 115.0  | 886.3  |
| Median balance (cents)   | 912.5  | 2822.5 |
| Min balance (cents)      | 527.0  | 880.0  |
| Max balance (cents)      | 1197.0 | 4335.0 |
| Mean trades per session  | 142    | 104    |
| Profit per trade (cents) | 7.44   | 52.73  |

The mean intraday balance progression in Figure 4 further illustrates this separation over time. The initial overlap of the two distributions persists up

to 1.73 hours after the market open, after which they separate and no longer intersect, indicating a distinct performance gap for the rest of the trading day. The wider shaded band for MMM02 reflects its considerably greater variance, which grows throughout the session from a standard deviation of 251 cents at 0.5 hours to 886 cents at close.



**Fig. 4.** Mean intraday balance progression over 30 trials. Shaded regions indicate the variation in the final balances given by  $\mu \pm \sigma$ .

## 5 Statistical Analysis

Before the statistical testing, the Shapiro-Wilk test was conducted to determine the normality of the final balances. It was concluded that MMM01 deviated significantly from normality ( $W=0.919$ ,  $p=0.026$ ), whilst MMM02 did not ( $W=0.972$ ,  $p=0.603$ ). Since one of the distributions was not normally distributed, non-parametric tests were used in conducting the statistical tests.

The Mann-Whitney U test was used to assess whether MMM02 balances were stochastically greater than MMM01 balances across independent trials, yielding  $U=881$  and  $p<0.0001$ , indicating a significant difference in final balance in favour of MMM02. Since both traders operated within the same market session, the observations are paired, making the Wilcoxon signed-rank test an appropriate complementary test, yielding  $W=464$  and  $p<0.0001$ , further confirming this result.

Cohen's  $d$  was computed to quantify the magnitude of the difference, yielding  $d = 2.979$ , signifying a large effect size under Cohen's guidelines [4]. This confirms that MMM02 performed significantly better than MMM01 across all trials.

## 6 Discussion and Conclusion

The results indicate that the MMM02 strategy outperforms MMM01 on all measures of performance, confirming that the three modifications address MMM01's

core weakness of fixed thresholds regardless of market conditions. MMM02 identifies uptrends using an EMA crossover, buys when the ask price falls sufficiently below the recent average based on market volatility, and limits inventory in downtrends. These modifications ensure that purchases are made more selectively and in favourable market conditions.

The profit per trade metric supports this interpretation. MMM02 executed fewer trades than MMM01 (104 vs 142 per session) yet had a substantially greater profit per trade (52.73 cents for MMM02 and 7.44 cents for MMM01). This can be attributed directly to two of the three modifications. Since the EMA crossover allows purchases only during uptrends, trades that MMM01 would have taken, due to its lack of trend awareness, are filtered out. Moreover, the volatility-adaptive threshold requires a higher discount when the market is more volatile, thereby increasing the entry criterion.

The intraday balance progression in Figure 4 reveals that the two distributions overlap in the early stages of the session until approximately 1.7 hours after market open, after which they diverge permanently. The initial overlap is consistent with the EMA signals requiring a warmup period to stabilise - with a slow EMA window of 51 trades, reliable trend signals cannot emerge until sufficient transaction history has accumulated. Once reliable trend signals emerge, MMM02 selectively enters favourable positions and maintains a consistently superior balance for the remainder of the trading day. Furthermore, MMM02's variance grows throughout the session - from a standard deviation of 251 cents at 0.5 hours to 886 cents at close - as trials detecting early uptrends diverge from those in flat or declining markets. This reflects an inherent trade-off between consistency and higher average returns, a natural consequence of a trend-following strategy that is more sensitive to market regime than the fixed-threshold approach of MMM01.

Another interesting observation concerns the Bayesian optimisation converging on a maximum inventory of  $K = 1$ . Despite the search range permitting values up to 5, single-unit holding was found optimal, implying the benefits of holding multiple units do not compensate for the associated risk. Halving the profit target to  $\delta_{\text{sell}}/2$  in downtrends encourages early exits, avoiding prolonged exposure to bearish markets as in MMM01.

MMM02 could be further improved in several ways. One method is applying a stop loss to provide explicit downside protection. MMM02 currently relies solely on downtrend detection to exit positions but has no hard limit on losses if the market moves sharply against it before a downtrend is detected. Additionally, incorporating more trend indicators such as MACD or RSI could help minimise false crossover signals in sideways markets, where price movements lack a clear trend and the simple EMA crossover may trigger unnecessary trades. On the other hand, use of too many indicators may result in overfitting and introduce conflicting signals, so any additions must be carefully validated. Time-of-day awareness would also be a valuable addition, since there are well-documented patterns of higher volatility at open and close that a fixed strategy cannot account for [5].

While MMM02 does demonstrate significant improvements over MMM01, several limitations should be acknowledged. First, the training and testing datasets only contain one day of trading each, meaning the hyperparameters may not generalise across different market conditions. Furthermore, BSE does not include some real-world microstructure features such as latency and market impact, implying performance in a live trading environment may differ from the results observed here. Finally, the 20 Bayesian optimisation trials may not have fully covered the parameter space, as evidenced by the best configuration being found in the final trial, indicating that additional trials could yield further improvements.

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